GinCol Lab · Aula

Student worksheet

Physical Education · 1st year of Bachillerato · Your body, your lab · Class 4 of 4 · With Isidora

The apps that measure you

Class video: https://youtu.be/CcQfuCNfZS4

The code leads to the video.

Name: _____ Date: _____

Before you start

1. Without looking at anything yet: if you or someone at home wears a sports watch or a fitness band, which of the numbers it gives do you trust most: pulse, steps or calories? And which do you trust least? Write it down: it's just for you.

2. **At the end of the class.** Go back to what you wrote. Did you get the order right? Why?

While you watch the video

1. **The green light.** What is on the underside of a sports watch? What exactly does the sensor measure: your pulse, or the light that comes back?

2. **Three ways.** What are the three ways each number on your watch can come from? Give an example of each one.

3. With each heartbeat, does more light come back, or less? Why?

4. Why is the light green and not red? Give the two reasons from the class.

5. **Is it right?** What is the reference measurement for the pulse, the one the watch is compared with? How is the comparison done?

6. At rest, on average, how far off is the watch? And each individual measurement? Why don't those two things contradict each other?

7. **When you move.** In the young woman climbing the stairs, which rhythm does the light at her wrist follow: her heart's or her movement's? How does the watch fix it?

8. Where is the watch's pulse furthest off: on the treadmill, on a bike or with weights?

9. For the oxygen in your blood, the watch does use red light. Why does red work there?

10. **It recognises: your steps.** What is inside an accelerometer? How does the watch decide that you have taken a step?

11. Give an example of something that makes it count too few, and another of something that makes it count too many.

12. **The modes.** On the treadmill, where does the watch get the distance from, without GPS? What happens if you hold on to the handrail?

13. With weights and in the pool, what movement does the watch look for to count a rep or a length?

14. On an exercise bike, what does the watch record? What does it need in order to know your cadence?

15. **It calculates: your calories.** What calorie sensor does the watch have? How are calories really measured, in a lab?

16. What data does the watch use to calculate your calories? How is that formula like "220 – age", from class 2?

17. Why, with weights, is the pulse a worse guide to the energy you use than when you are running?

18. **Your sleep.** In a sleep lab, where are the stages read? And what does the watch look at? At what moment of the night is it furthest off?

19. **Your data.** What could be seen on Strava's map in 2018, if nobody had given their name? What do your routes say about you, if they always start and end at the same place?

20. Why is your pulse, stored day after day, health data? And what about the calories that the watch gets wrong?

21. In Spain, from what age do you give permission yourself for an app to use your data?

Does it measure it, recognise it or calculate it?

Mark **M** if the watch **measures** it with a sensor (even if it processes it afterwards), **R** if it **recognises** it by looking for a shape in your movement, and **C** if it **calculates** it with a formula or a program made by measuring other people.

1. How much green light comes back from your wrist.

2. Your calories from a Physical Education class.

3. The lengths you have swum.

4. The jolts of your wrist.

5. The reps in a set of weights.

6. Your sleep stages.

7. The metres you have climbed on a staircase.

8. Your "fitness age" or "fitness level", which some apps give you.

9. Look on your watch, or on that of someone at home, for a number that isn't in the class. Which family is it in? How do you know?

Practice

1. **Reliable or valid.** Three mornings in a row, sitting down, you count your pulse for a minute and get 80 each time; your watch shows 84 all three times. For you, is it reliable? Is it valid? By what percentage does it overshoot?

2. **Green or red.** Explain in your own words why the watch measures your pulse with green light and not red.

3. **Permissions.** A step-counting app asks for: your location all the time, your contacts, the microphone and your physical activity. Which ones does it really need to count your steps? And if you also want to see the map of your routes?

Put it to the test, if you feel like it

All at home, with whatever you have. If something isn't right, stop. If you don't have a watch or an app, that's fine: what you would see, you have already seen in the video.

Test	You	The device
Your pulse, sitting down: two fingers on your wrist, a whole minute		

Test	You	The device
A hundred steps around the house, counted, with your phone in your pocket	100	

And the permissions of your sports app: on an iPhone, in Settings, the app, "Motion & Fitness" and "Location"; on Android, in Settings, Apps, the app and "Permissions" (for location, better "Allow only while using the app"). Remove the ones it doesn't need.

Keep thinking

Who would you let see your watch data: your family, your school, a health insurer? Why some and not others? If you like, share it in the class forum.

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